

Job Simulation: User Table with Fake API & DOM Manipulation

Project Title: Implementation of a User Management Table using JSON API and DOM

Role: Front-End Development Intern

Technology Stack: HTML, CSS, JavaScript

Objective

To design and implement a dynamic user table that fetches data from a fake JSON API and displays it using DOM manipulation. This table will allow editing and deleting rows directly from the browser, providing hands-on experience in working with the Fetch API and real-time UI updates without using any frameworks or localStorage.

Task Overview

As part of your hands-on learning, you are required to fetch user data from a JSON API (e.g., <https://jsonplaceholder.typicode.com/users>) and render it in an HTML table. The table should include **Edit** and **Delete** buttons for each row. When a user clicks **Edit**, the row content becomes editable. When they click **Delete**, the row is removed from the table.

The table must be styled using CSS with readable fonts and colors for better UI presentation.

Task Requirements

1. Functionality:

- Use the Fetch API to retrieve user data from a fake JSON endpoint
- Dynamically generate an HTML table using JavaScript DOM methods
- Add two action columns for each row: **Edit** and **Delete**
- Edit button should allow the user to modify the row content inline
- Delete button should remove the row from the DOM

2. User Interface (UI):

- Apply styling using CSS for headers, rows, and alternating background colors
- Use visually distinct buttons for Edit (e.g., blue) and Delete (e.g., red)
- Ensure table alignment and spacing are consistent

3. Code Structure:

- Use `document.createElement()` and `appendChild()` for building the DOM
- Use `addEventListener()` to attach actions to Edit and Delete buttons
- Format text clearly and maintain clean modular code with functions

Bonus (Optional)

- Display additional user details such as username or phone
- Use different background colors for even/odd rows
- Add a hover effect on table rows
- Disable editing when the user clicks “Save” again (toggle Edit/Save)

Deliverables

Submit a project folder that includes:

- `index.html` – HTML structure for the table and container.
- `style.css` – Styling rules for table, buttons, and layout
- `script.js` – JavaScript logic to fetch data and implement DOM CRUD actions

Note: Include a brief (2–3 line) comment in `script.js` explaining how the fetch and edit/delete logic works.

Learning Outcomes

By completing this task, you will gain practical experience in:

- Fetching JSON data from a REST API using `fetch()`
- Dynamically creating and updating HTML elements using the DOM
- Implementing Edit/Delete functionality with event listeners
- Structuring and styling responsive tables using CSS